

Devansh Srivastava

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EDUCATION

IIIT Naya Raipur

Bachelor of Technology in Data Science and Artificial Intelligence
CGPA: 9.27

Naya Raipur, Chhattisgarh

November 2022 – July 2026

Relevant Courses

Linear Algebra, Calculus III, DSA, Operating Systems, DBMS, Object-Oriented Analysis and Design, Deep Learning, Computer Vision, Blockchain, and Natural Language Processing.

EXPERIENCE

Data Engineer Intern: Karya Inc. | PostgreSQL, Pandas, Bash

November 2025 | [Karya](#)

- Developed and optimized automated ETL pipelines using Bash and Python scripts, facilitating seamless data ingestion and processing from diverse sources.
- Collaborated with cross-functional teams to gather data requirements, designing scalable ETL solutions that enhance data accessibility and usability.

Summer Intern: IIT Ropar | Node.js, MongoDB, Cloudinary, YouTube Data API, Render May 2025 | [report](#)

- Designed and deployed secure JWT-authenticated APIs with scalable MongoDB schemas and automated YouTube Data API ingestion for 1K+ videos.
- Automated deployment pipelines for backend (Render) and frontend (Vercel), achieving 100% resolution of production issues and reducing release time.

ACHIEVEMENTS

- Secured **AIR 84** in **Amazon ML Challenge** out of **20,000+ teams** with a SMAPE score of 43.137. [🔗](#)
- Selected as **one of 3,000 students out of 17,500 applicants** for **Amazon ML Summer School India**. [🔗](#)
- Selected as one of 119 **global contributors** in the **Passagemath Organization** collaborating on computational mathematics tools used by researchers worldwide (SageMath fork). [🔗](#)
- Won AIML track at **Hack-o-Harbour Hackathon** among 40+ teams.
- Secured **2nd** place at **Industry Academia Meet Ideathon 2025**, competing against 50+ teams.

OPEN SOURCE CONTRIBUTIONS

Open Source Contributor: Sage Math| Python, Cython, C

[github](#)

- Introduced hashability support for the **Combination class** enabling O(1) lookups. [🔗](#)
- Automated a version validation to reject **FriCAS** version $\leq 1.3.7$. [🔗](#)
- Resolved a critical bug in SageMath's logarithm method to handle non-positive bases in the **Complex Field**, increasing mathematical computation accuracy by 50%. [🔗](#)

PROJECTS

PerfLens | Rust, eBPF, libbpf, C, Ratatui, HuggingFace API, Tokio

[WIP] | [github](#)

- Built eBPF monitor tracking slow I/O (<5ms) with AI root cause analysis and stack traces.
- Integrated HuggingFace LLM for bottleneck detection; visualized process fork/exec chains.

Carrot | C, Cmake, Event Loops, Data Structures, Docker

[WIP] | [github](#)

- Engineered a **Redis-like in-memory key-value** database from scratch (~4,800 LOC) that supports both basic KV operations and sorted sets.
- Executed an **event loop** using UNIX's poll system for managing concurrent request handling. Integrated AVL trees for implementing sorted sets and ttl caching.

Carbon-Credits | Ethereum, Sepolia API, Hardhat, EthersJs, Flask, ReactJs, PostgreSQL December 2024 | [github](#)

- Designed and deployed a decentralized Carbon Credit trading platform leveraging React, Flask, and Solidity, promoting transparency and accountability in CSR activities and integrated a royalty mechanism ensuring 10% of transaction value is returned to NGOs.
- Built a web interface that implements a decentralized carbon market, issuing 100% verifiable proof certificates for proof of purchase.

TECHNICAL SKILLS

Languages: C, C++, Python, Rust, C#, JavaScript, TypeScript

Tools: MySQL + PostgreSQL, Bash, Numpy, Pandas, Scikit-learn, Make, CMake, TensorFlow, HuggingFace, OpenCV, PyTorch, Flask, FastAPI, MongoDB, ReactJs

Technologies: REST API Development, GNU/Linux, Git, GitHub, SQL/NoSQL databases, Vim/Neovim, Prometheus, Distributed Systems, Kubernetes, NodeJs, ExpressJs, Event-driven Systems, Docker, Kafka, Grafana, GCP, AWS, Redis, L^AT_EX